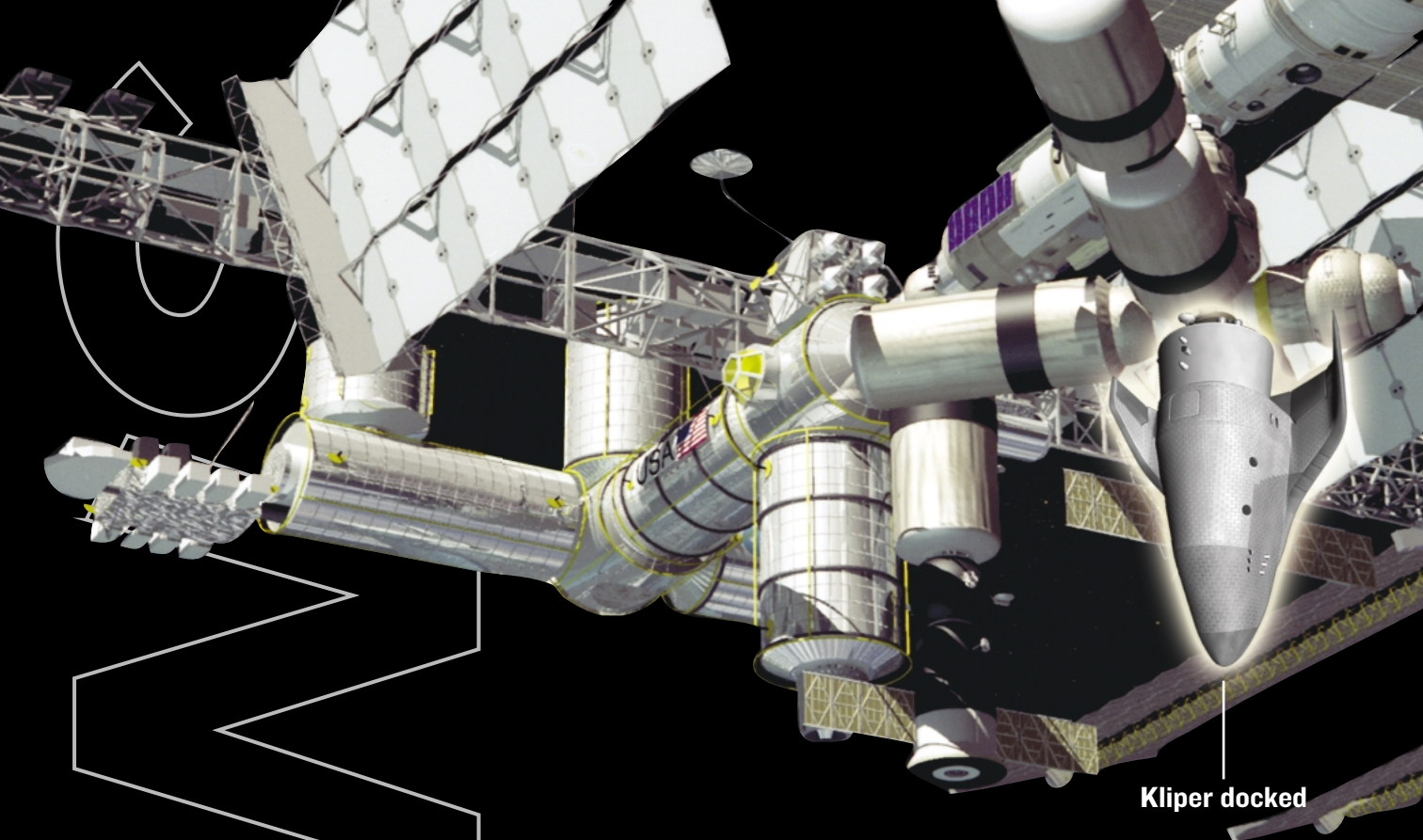




Kliper docked

Europe to Join Russia in Building Next Space Shuttle

Development agreement takes shape during the Paris Air Show

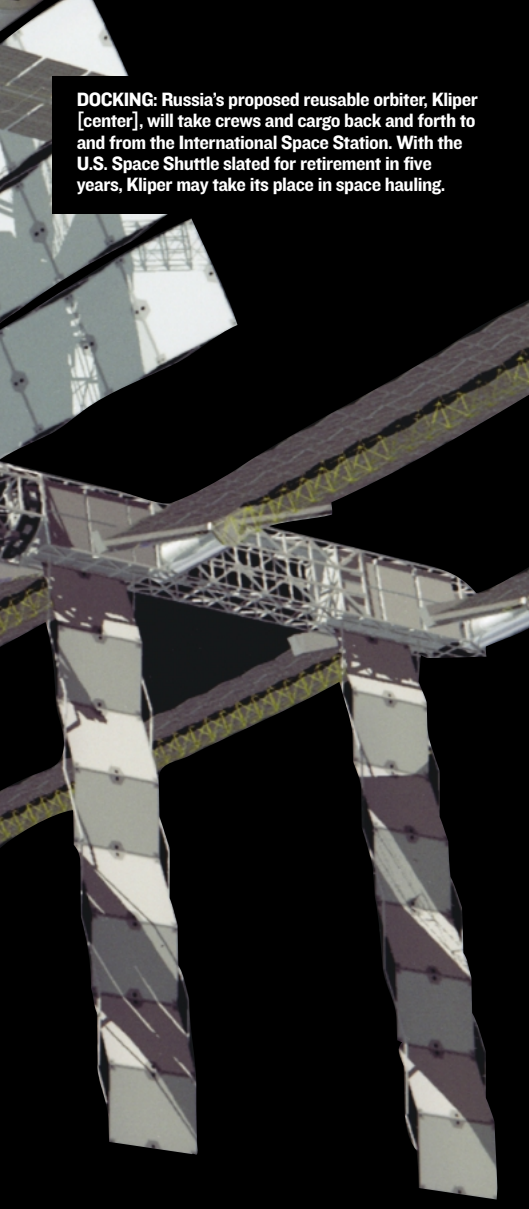
It's all but official—Russia and Europe will soon embark on a cooperative effort to build a next-generation manned space shuttle. Speaking at the Paris Air Show, in Le Bourget, France, in June, Russian space officials confirmed earlier reports from Moscow that their partners at the European Space Agency would join the Russian effort to build a new reusable orbiter, dubbed Kliper. After the cautious optimism they expressed at the beginning of 2005, Russians are now confident that their European partners will be on board for the largest, boldest Russian endeavor in spaceflight in more than a decade.

"The prospects of our cooperation [with Europe] are very good, and I think we [soon] will see official statements to this effect," said Alexander Derechin, the head of international relations at S.P. Korolev Rocket and Space Corp. Energia (known as RKK), in Korolev, Russia's prime developer of manned spacecraft. The question now is what contributions

in hardware and expertise the European partners will provide.

According to Vladimir Daneev, head of the division responsible for the development of Kliper at RKK, his company will likely seek European expertise in the development of the spacecraft's crew module, onboard avionics, and thermal protection. "We now want to understand which available European and Russian systems are best suited for this project," Daneev said.

THE RUSSIANS, encouraged by support from their prospective foreign partners, came to Le Bourget in June with a better, bolder Kliper than the one they were showing off just a few months ago. Instead of the wingless vehicle originally envisioned [see News, "Russians Propose a New Space Shuttle," *IEEE Spectrum*, February], RKK now favors a minishuttle with swept wings. It will double the cargo and crew capacity of



DOCKING: Russia's proposed reusable orbiter, Kliper [center], will take crews and cargo back and forth to and from the International Space Station. With the U.S. Space Shuttle slated for retirement in five years, Kliper may take its place in space hauling.

the Soyuz capsule, replacing its venerable predecessor, which served more than four decades as the transport and resupply vehicle for the International Space Station (ISS).

"Today, we see a winged version as more promising," says Alexander Safonov, deputy head of the Department of Transport Space Systems at RKK, "because it can land at any Class I airport with a runway length of 2.5 to 3.5 kilometers." Wings will enable the spacecraft to maneuver as far as 1500 km from its entry path during landing to reach its destination.

Kliper's principal mission will be to ensure that the ISS is effectively supplied and maintained, after the planned retirement of the U.S. Space Shuttle [see illustration, "Docking"]. But Russian officials hope it might also become the cornerstone of a future deep-space exploration program, which would parallel NASA's efforts to return to the moon.

Russian-European cooperation related to the ISS has included RKK's supplying of critical parts

AIRBUS VS. BOEING: SOME BATTLE STATS

THE BIGGEST SPLASH at the Paris Air Show in June was made by Airbus's A380 super-jumbojet, though its maiden flight was a few months earlier. The massive plane, designed to carry 555 to 800 passengers, wowed onlookers, who were amazed at how quiet it was.

But all was not quiet at Airbus's Toulouse, France, headquarters. Instead of emerging dominant after outselling its arch-rival Boeing in world markets for four of the last five years, Airbus may have made missteps that let Boeing, now headquartered in Chicago, back into the game.

The A380, intended to be Airbus's flagship and to displace the Boeing 747 as the world's long-haul carrier of choice, has cost €3 billion more than intended; is six months behind schedule; and, despite designers' best efforts, is still way too heavy to land at most airports without damaging runways or endangering passengers.

As recently as a year ago, it wasn't clear that Boeing would ever build its 787 Dreamliner, the midsize jet it was envisioning as its new standard-bearer. But advance sales of the sleek new 300-passenger jet have been brisk. Airbus, now worried about abandoning the medium-haul market to Boeing, petitioned the European Union for money to design and build a direct competitor, the A350.

That sparked suits and countersuits at the World Trade Organization, with the U.S. government saying that European subsidies are illegal and the EU countering that Boeing's military contracts and tax breaks amount to the same thing.

—WILLIE D. JONES

0

Boeing 787 Dreamliner orders as of 31 March 2004

266

787 orders as of 30 June 2005

129

Airbus A380 superjumbo orders as of 31 March 2004

159

A380 orders as of 30 June 2005

\$120 million

787 list price in U.S. dollars

\$295 million

A380 list price in U.S. dollars

12

Number of airports that can accommodate the A380

60

Number of airports being upgraded for the A380

355 000

Boeing 747 maximum takeoff weight in kilograms

560 000

A380 maximum takeoff weight in kilograms

125

Airbus midsize A350 orders placed at Paris Air Show

0

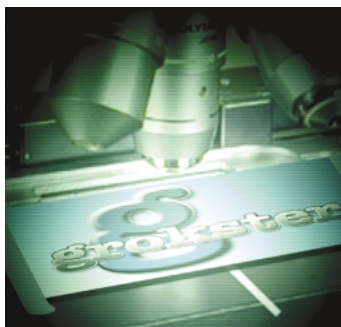
787 orders placed at Paris Air Show

Sources: AirbusA380.com, Boeing Co., BusinessWeek, CNN/Money, Dow Jones & Co., Miami Herald, SpeedNews.com, USA Today.

NEWS BRIEFS

FUSION SOLUTION. After a years-long standoff between Japan and the European Union over where the world's next nuclear fusion reactor would be sited, the consortium looking to build the 500-megawatt Iter reactor voted on 28 June to place the €10 billion (US \$12 billion) machine in Cadarache, France. The reactor will demonstrate sustained plasma burning, to establish the foundation for a commercial prototype plant. The Japanese obtained a promise that they will be well represented in the project, and the head of the Iter organization will be Tokyo's choice.

NOT FAIR TO SHARE. On 27 June, the U.S. Supreme Court overturned a lower court ruling that said companies running peer-to-peer file-sharing networks cannot be held liable for copyright infringement as long as the networks can be used for legitimate purposes as well. The decision makes it easier for copyright holders such as music labels and movie studios



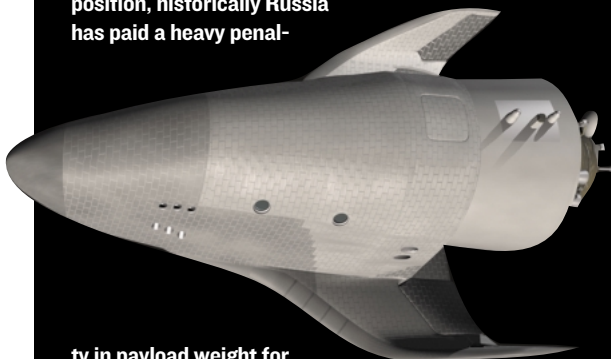
to sue file-sharing networks such as Grokster [see image] and Morpheus—firms that had escaped liability because they do not house files on central servers.

SEOUL SINKS SCIENCE. Proposed changes to South Korea's national educational curriculum suggested by the Ministry of Education will make science courses optional for students in the final two years of high school. The agency explains the seemingly odd move as a way for students to customize their learning experiences, to make them best suited to the career

of the European Space Agency's *Jules Verne* cargo ship. The *Jules Verne*, however, has a major drawback: it can't bring cargo back from the station—a capability that will become critically important as the U.S. Space Shuttle's retirement looms, around 2010. Having invested more than two decades in building a scientific laboratory for the ISS that has yet to be launched, ESA is keen to have a reliable two-way supply line to the ISS.

Conveniently, in the middle of 2004 a Russian proposal came for cooperation in building Kliper. Initially, ESA was interested in an unmanned wingless orbiter, which would be launched on top of the European Ariane-5 rocket. The Russians, however, insisted on a full-scale piloted vehicle. ESA delegated the European Aeronautic Defense and Space Co., the aerospace consortium, to study the proposal.

COOPERATION WITH EUROPE promises to give Russia unprecedented flexibility in its access to orbit. Bound by its geographic position, historically Russia has paid a heavy penal-



ty in payload weight for launching its spacecraft from such sites as Baikonur or Plesetsk, both far from the equator. Today, RKK is seriously considering either launching Kliper on top of the existing Zenit rocket from a floating platform in the Pacific Ocean or on upgrading the Soyuz rocket, which could fly from the equatorial site in Kourou, French Guiana.

The ink is barely dry on an agreement between Europe and Russia to build a launchpad for the Soyuz rocket in Kourou. The contract, signed on 26 April, calls for the completion of the pad within 35 months. Igor Barmin, head of KBOM, a main developer in Moscow of launch systems, confirmed at Le Bourget that European funding for the Soyuz pad in Kourou has started flowing. Added construction will be required to accommodate the Soyuz-3 rocket that would be capable of lifting Kliper. (Barmin is the son of the legendary space pioneer Vladimir Pavlovich Barmin, who built the Baikonur Cosmodrome in Kazakhstan and designed many of the Soviet Union's rocket launch systems.)

While the completion of the new Soyuz pad in Kourou is three years away, RKK will likely have to make a decision on the launcher much sooner. "Both launchers, Zenit and Soyuz-3, have to be upgraded for launching manned spacecraft, and we might have to choose the one on which we want to spend money," says RKK's Derechin.

After opting for the winged shuttle design, the RKK leadership persuaded OKB Sukhoi, the Moscow military contractor best known for development of the Su-27 jet fighter, to invest its own resources and expertise in the Kliper project. RKK officials admitted that OKB Sukhoi's solid financial standing, along with the company's unmatched experience in supersonic and subsonic aerodynamics, led them to seek out this partnership. Their alternative would have been to join with the financially strapped NPO Molniya, their Moscow-based partner in the development of the Buran orbiter, which was to have been the Soviet counterpart to the U.S. Space Shuttle.

AS FOR RELATIONS WITH NASA, after more than a decade of cooperation, the Russians had few illusions. Painful Russian delays in the delivery of key components for the space station during the 1990s had prompted the U.S. Congress and the press to pound NASA for "letting Russia

COOPERATION WITH EUROPE PROMISES TO GIVE RUSSIA UNPRECEDENTED FLEXIBILITY IN ITS ACCESS TO ORBIT

into the critical path of the station," to use a favored expression. In 2000, Congress prohibited NASA from buying Russian hardware and services altogether, because of Russia's nuclear and missile cooperation with Iran. This meant, among other things, that Russia could not bid on contracts for NASA's Crew Exploration Vehicle (CEV), planned to take astronauts to the moon and Mars and to put them into orbit.

"There was no question but that NASA was going to build the CEV on its own," says Derechin. "America would not accept any dependency on Russia." Therefore, any interaction between the Kliper and CEV projects will be limited to "programmatic and engineering compatibility," he says. "We don't want a repetition of the Soyuz-Apollo [docking mission], where we had air and they had [pure] oxygen [in the life-support system]." (Incompatibility of the atmospheres of the two spacecraft seriously complicated transfers between them.) The CEV is intended to enter service after 2010, when the current Space Shuttle fleet is retired, and Kliper is expected to follow suit within two years.

The Russians and Europeans, with few good alternatives, have seen their aspirations and needs converge around the Kliper project. If they are successful, this could open new horizons for manned spaceflight in this century. —ANATOLY ZAK